

Appendix A — Mathematical Formalism of Phase Theory

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This document constitutes Appendix A of the above-titled preprint.

This appendix provides the complete symbolic infrastructure underlying Phase Theory. All structures are built from the ground up: no spacetime manifold, no pre-existing metric, and no quantum or classical fields are assumed as primitives. Every familiar physical object — particles, spacetime geometry, causal order, gauge interactions, and gravity — is derived as an emergent feature of a single globally defined phase-bearing substrate. The notation is chosen to be self-contained; cross-references to the main text are given in brackets of the form [§n].

A. THE FUNDAMENTAL OBJECT

Definition A.1 (Phase Configuration). Let C be a *phase configuration space* — a separable, metrizable topological space equipped with the following three structures:

1. **Continuity structure.** A topology τ_C such that phase configurations vary continuously with respect to a substrate parameter u (defined in §C below).
2. **Admissible topology class.** A distinguished class $[C]_{top}$ of allowable global topologies, which restricts phase configurations to those realizing homotopy classes compatible with finite inconsistency (§B).

3. Coherence bounds. A pair $(\rho_{min}, \rho_{max}) \in \mathbb{R}_{>0}^2$ such that the local coherence density $\rho_{coh}(u)$ satisfies $\rho_{min} \leq \rho_{coh} \leq \rho_{max}$ everywhere.

A phase configuration is a map:

$$\Phi : C \rightarrow S^1 \times H_{top}$$

where $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ is the circle (encoding local phase angle $\theta \in [0, 2\pi)$) and H_{top} is the space of admissible holonomy structures.

The local phase at a substrate point $p \in C$ is written $\theta(p) \in [0, 2\pi)$, and the associated connection proxy is:

$$A(p) \in T^*C \otimes \mathfrak{g}$$

where $\mathfrak{g}$ is the Lie algebra of the admissible holonomy group G .

Remark A.1. C is emphatically *not* a spacetime manifold. It carries no metric at this stage; the substrate measure $d\mu$ introduced in §B is not a volume form derived from any metric. Spacetime geometry emerges as a derived structure in §H.

B. THE GLOBAL PHASE-INCONSISTENCY FUNCTIONAL

Definition B.1 (Inconsistency Functional). For a phase configuration $\Phi \in C$, define the *global phase-inconsistency functional*:

$$I[\Phi] = \int_C (\alpha |\nabla \theta|^2 + \beta |\nabla \times A|^2 + \gamma T(\Phi) + \delta C_{coh}(\Phi)) d\mu$$

Table B.1. — Terms of the Inconsistency Functional $I[\Phi]$

Term	Symbol	Physical Role
Phase gradient energy	$\alpha \nabla \theta ^2 \geq 0$	Penalizes rapid local phase variation
Curvature / holonomy energy	$\beta \nabla \times A ^2 \geq 0$	Penalizes holonomy inconsistency (proto-gauge curvature)
Topological obstruction	$\gamma T(\Phi) \geq 0$	Penalizes configurations

Term	Symbol	Physical Role
		with non-trivial but energetically unsupported topology
Coherence penalty	$\delta C_{coh}(\Phi) \geq 0$	Penalizes configurations that violate coherence bounds

The coupling constants $\alpha, \beta, \gamma, \delta > 0$ are fundamental parameters of Phase Theory, fixed by global consistency self-referentially — see §B.3 of main text.

Definition B.2 (Topological Obstruction Term). Let $\pi_n(C)$ denote the n -th homotopy group of C . The topological obstruction term is:

$$T(\Phi) = \sum_{n \geq 1} w_n \cdot O_n(\Phi)$$

where $O_n(\Phi) \geq 0$ measures the degree to which Φ deviates from the unique ground-topology representative in homotopy class $[\Phi]_n$, and w_n are monotonically decreasing weights encoding the relative energy cost of higher-order topological defects.

Definition B.3 (Coherence Penalty). Define the local coherence density:

$$\rho_{coh}(p) = \langle \exp(i[\theta(p) - \theta(p')]) \rangle_{p' \in B_\epsilon(p)}$$

where the average is taken over a substrate neighborhood $B_\epsilon(p)$ of p . Then:

$$C_{coh}(\Phi) = \int_C \max(0, \rho_{min} - \rho_{coh}(p))^2 d\mu(p)$$

Proposition B.1 (Non-negativity and coercivity). $I[\Phi] \geq 0$ for all $\Phi \in C$, and $I[\Phi] = 0$ only for the unique (up to global gauge) ground-state configuration Φ_0 of trivial topological class.

Proof. Each integrand term is a sum of squared non-negative quantities under appropriate norms. Uniqueness of the zero follows from strict convexity of the coherence penalty in the neighborhood of perfectly coherent configurations. $\square$

C. EVOLUTION LAW — DYNAMICS IN PRE-TIME

Definition C.1 (Ordering Parameter). Let $u \in \mathbb{R}$ be a real-valued *ordering parameter* — a coordinatization of the global sequence of substrate updates. u is not identified with physical time at this stage.

Evolution Equation

The dynamics of Φ are governed by:

$$\partial\Phi/\partial u = -\delta I/\delta\Phi + \sum_i \lambda_i(u) K_i(\Phi)$$

where:

- $\delta I/\delta\Phi$ is the functional (Fréchet) derivative of I with respect to Φ , driving the configuration toward lower inconsistency.
- $K_i(\Phi)$ are constraint functionals enforcing: (a) *Topological rigidity*: homotopy class preserved unless a topological transition event occurs. (b) *Coherence bounds*: $\rho_{coh}(p) \in [\rho_{min}, \rho_{max}]$ pointwise. (c) *Normalization*: global phase invariance $\Phi \sim e^{i\psi} \Phi$ for constant ψ .
- $\lambda_i(u)$ are Lagrange multipliers determined self-consistently by the constraint equations $K_i(\Phi(u)) = 0$.

Remark C.1. This equation has the form of a constrained L^2 -gradient flow on $\mathcal{C}$. In finite-dimensional analogy, it is the projection of steepest descent onto the constraint manifold defined by the K_i . The key distinction from standard variational mechanics is that u is *pre-temporal* — it provides ordering without duration.

Proposition C.2 (Monotone decrease of inconsistency). Along any solution $\Phi(u)$ of the evolution equation:

$$dI[\Phi(u)]/du \leq 0$$

with equality only at stationary points $\delta I/\delta\Phi = 0$ (stable defects).

Proof. By the chain rule:

$$dI/du = \langle \delta I / \delta \Phi, \partial \Phi / \partial u \rangle = \langle \delta I / \delta \Phi, -\delta I / \delta \Phi + \sum_i \lambda_i K_i \rangle.$$

Orthogonality of the constraint directions K_i to the inconsistency gradient (by the KKT conditions at each u) gives $\langle \delta I / \delta \Phi, K_i \rangle = 0$, yielding $dI/du = -\|\delta I / \delta \Phi\|^2 \leq 0$. $\square$

D. EMERGENT TIME

Definition D.1 (Update Density). For a solution $\Phi(u)$, define the *update density* at substrate point p as:

$$\rho_{\text{update}}(p, u) = |\partial \theta(p, u) / \partial u|$$

Definition D.2 (Physical Time). For a substrate trajectory Γ_x passing through a sequence of substrate events associated with an observer x , define physical time by:

$$t(x) \propto \int_{\Gamma_x} \rho_{\text{update}}(u) du$$

The proportionality constant is fixed by normalization to SI units via a reference oscillator.

Theorem D.1 (Emergent time dilation). Let Γ_1, Γ_2 be two observer trajectories in C with identical ordering-parameter endpoints $[u_a, u_b]$. If Γ_2 passes through a region of elevated phase-gradient energy, then:

$$t(\Gamma_1) \neq t(\Gamma_2)$$

In the weak-strain limit (§H), the ratio $t(\Gamma_2)/t(\Gamma_1)$ reproduces the gravitational time dilation formula of general relativity.

Proof. The update density ρ_{update} is suppressed in high-inconsistency regions (where $\|\nabla \theta\|^2$ is large), since the local relaxation rate is retarded by the increased restoring force. Quantitative agreement with GR follows from the linearization of $g^{\text{eff}}_{\mu\nu}$ in §H. $\square$

Corollary D.2. The arrow of time is not postulated but derived: the ordering parameter u provides global directionality, and physical time inherits this directionality from the monotone decrease of I (Proposition C.2).

E. PARTICLE SECTORS — TOPOLOGICAL DEFECT QUANTUM THEORY (TDQT)

Definition E.1 (Topological Sector). A particle species in Phase Theory is a stable, localized topological defect of Φ , classified by an element of a homotopy group of C :

$$\pi_n \in \pi_n(C), n \geq 1$$

The sector index n determines the codimension of the defect in C .

Definition E.2 (Stability Conditions). A defect $\pi_n(\Phi)$ is stable (corresponds to a physical particle) if and only if:

$$\delta I / \delta \Phi|_{\Phi=\Phi_{\text{defect}}} = 0 \quad (\text{stationarity})$$

$$\delta^2 I / \delta \Phi^2|_{\Phi=\Phi_{\text{defect}}} > 0 \quad (\text{local minimality})$$

i.e., the defect configuration is a strict local minimum of I within its homotopy class.

Definition E.3 (Particle Label Set). Each stable defect is uniquely characterized by the quintuple:

$$p = (k, Q_H, t, \chi, R)$$

Table E.1. — Particle Label Quintuple

Label	Name	Definition
$k \in \mathbb{Z}$	Phase winding number	Degree of the map $\Phi _{\partial D^2} : S^1 \rightarrow S^1$
$Q_H \in \mathbb{Z}$	Hopf invariant	$Q_H = \int A \wedge dA$, with A the Hopf form
$t \in \mathbb{Z}_{\geq 0}$	Excitation index	Radial mode number of the defect profile
$\chi \in \{+1, -1\}$	Framing / chirality	Orientation of the defect's framing bundle
R	Holonomy representation	Irreducible representation of G carried by the defect

Definition E.4 (Mass Functional). The mass of a particle defect p is defined as:

$$m[p] = I[\Phi_p] - I[\Phi_0]$$

the excess inconsistency relative to the ground state. This is finite by the stability conditions of Definition E.2 and non-negative by Proposition B.1.

Theorem E.5 (Finite spectrum). Under the admissible topology conditions of Definition A.1, the number of distinct stable topological sectors satisfying Definitions E.2–E.3 is finite, bounded above by:

$$N_{max} \approx 25 \pm 10$$

Proof sketch. The admissible topology class $[C]_{top}$ restricts the available homotopy groups. The coherence bounds (ρ_{min}, ρ_{max}) impose an ultraviolet cutoff on viable defect profiles, excluding configurations with divergent excitation indices. The Hopf invariant is bounded by the global holonomy budget of I . The precise count requires numerical minimization of I over the discrete label set (see main text §VII). $\square$

F. ANTIMATTER — CONJUGATION SYMMETRY

Definition F.1 (Phase Conjugation). Define the *conjugation operator*:

$$C : \Phi(k) \mapsto \Phi(-k)$$

acting by reversal of the winding number $k \rightarrow -k$ while leaving Q_H , t , $|\chi|$, and R invariant (with $\chi \rightarrow -\chi$ when parity is broken in the defect sector).

Proposition F.1 (Mass-conjugation invariance). The mass functional is invariant under phase conjugation:

$$m[\Phi(k)] = m[\Phi(-k)]$$

Proof. Since I depends on θ only through $|\nabla\theta|^2$ and $|\nabla \times A|^2$ — both even in k — the total inconsistency is symmetric under $k \rightarrow -k$. $\square$

Proposition F.2 (Gravitational equivalence of matter and antimatter).

Because $m[\Phi(k)] = m[\Phi(-k)]$ and the emergent gravitational coupling depends only on $m[p]$ (see §H), matter and antimatter defects couple to emergent geometry identically. Phase Theory predicts no gravitational repulsion between matter and antimatter — a falsifiable consequence distinguishing it from certain competing proposals.

G. ANNIHILATION — CONJUGATE ANNIHILATION RELAXATION THEORY (CART)

Definition G.1 (Conjugate Pair). Let $\Phi_1 = \Phi(k, Q_H, t, \chi, R)$ and $\Phi_2 = C(\Phi_1) = \Phi(-k, Q_H, t, -\chi, \bar{R})$ be a conjugate defect pair.

Definition G.2 (Annihilation Pathway). An *annihilation channel* is a sequence of topological transitions:

$$\Phi_1 \oplus \Phi_2 \rightarrow \{\Phi_{meson}\} \rightarrow \Phi_{rad}$$

where:

- $\Phi_1 \oplus \Phi_2$ denotes the composite configuration formed by bringing the two defects within interaction range (overlap of their inconsistency halos).
- $\{\Phi_{meson}\}$ is the set of intermediate bound or resonant defect states (carrying the sum topological charge, which is zero for conjugate pairs).
- Φ_{rad} is the final radiation configuration — a coherent propagating phase mode (photon analogue, §K).

Topological Selection Rule

The allowed annihilation channels satisfy the inconsistency-descent condition:

$$\Delta I_{allowed\ channel} < \Delta I_{direct\ relaxation}$$

i.e., the pathway through intermediate meson-like defects achieves a lower effective inconsistency barrier than direct collapse. This condition determines the branching ratios of annihilation products [§VIII of main text].

Proposition G.3 (Topology-dependent channel selection). The Hopf invariant Q_H and holonomy representation R of the annihilating pair determine which intermediate states $\{\Phi_{meson}\}$ are topologically accessible. Pairs with $Q_H \neq 0$ are forbidden from direct two-radiation annihilation, predicting anomalous multi-radiation final states as a falsifiable signature.

H. EMERGENT GEOMETRY AND GRAVITY

Definition H.1 (Transport Metric). Given a phase configuration Φ with local phase gradient $\nabla\Phi$, define the effective transport metric:

$$g_{\mu\nu}^{eff} = G_{\mu\nu}(\nabla\Phi)$$

where $G_{\mu\nu}$ is the metric functional, a symmetric positive-definite bilinear form derived from the second-order Taylor expansion of I about the defect-free ground state. Explicitly, in the weak-strain limit:

$$g_{\mu\nu}^{eff} \approx \eta_{\mu\nu} + h_{\mu\nu}(\nabla^2\theta, \nabla \times A)$$

where $\eta_{\mu\nu}$ is the flat Minkowski tensor and $h_{\mu\nu}$ is the linearized strain perturbation — recovering GR linearized gravity (§K).

Definition H.2 (Refractive Index). Define the local effective refractive index of the phase substrate as:

$$n(x) = f(\rho_{strain}(x))$$

where $\rho_{strain}(x) = |\nabla\theta(x)|^2$ is the local phase-gradient energy density, and f is a smooth, monotonically increasing function with $f(0) = 1$ (flat substrate limit).

Definition H.3 (Geodesic Replacement). Classical trajectories are replaced by *least-inconsistency transport paths* — curves $\gamma : [u_1, u_2] \rightarrow C$ minimizing the path inconsistency functional:

$$I_{\text{path}}[\gamma] = \int_{u_1}^{u_2} n(\gamma(u)) |\dot{\gamma}(u)| du$$

Theorem H.4 (Gravitational lensing from refractive index). In the vicinity of a massive defect (high- I localized region), the refractive index $n(x) > 1$ causes least-inconsistency paths to curve toward the defect. In the weak-strain limit, the deflection angle of a radiation mode reproduces the GR prediction:

$$\delta\varphi = 4GM/c^2 b$$

(for impact parameter b), with corrections of order $O(V^2 \theta^2)$ that constitute a falsifiable deviation from GR [Prediction Class VI, main text §IX].

I. THE CAUSALITY CONSTRAINT — NO CLOSED PHASE LOOP THEOREM (NCPLT)

Definition I.1 (Update Ordering Graph). Let G_u be the directed graph whose vertices are substrate states $\{\Phi_i\}$ at successive values of u , and whose directed edges encode the update relation: $\Phi_i \rightarrow \Phi_j$ iff Φ_j is the immediate successor of Φ_i under the evolution equation of §C.

Theorem I.2 (No Closed Causal Loops — NCPLT). The ordering graph G_u contains no closed directed cycles:

$$\nexists \text{ closed directed cycle in } G_u$$

Proof. By Proposition C.2, $I[\Phi(u)]$ is monotonically non-increasing in u . A closed directed cycle $\Phi_0 \rightarrow \Phi_1 \rightarrow \dots \rightarrow \Phi_n \rightarrow \Phi_0$ would require:

$$I[\Phi_0] \geq I[\Phi_1] \geq \dots \geq I[\Phi_n] \geq I[\Phi_0]$$

forcing I to be constant along the cycle. By strict monotonicity off fixed points, this forces $\delta I / \delta \Phi = 0$ everywhere on the cycle — contradicting the assumption of a non-stationary trajectory. $\square$

Corollary I.3 (Prohibition of time travel). Since physical time $t(\S D)$ is a monotone functional of the ordering parameter u along any observer trajectory, and G_u is acyclic, no physical observer can return to an earlier t -value. Causal paradoxes are forbidden by construction in Phase Theory — not as a postulate, but as a theorem.

J. BLACK HOLE ANALOGUE — GRADIENT SATURATION

Definition J.1 (Saturation Condition). A region $R \subset C$ is said to be *gradient-saturated* if:

$$|\nabla \Phi(p)| \rightarrow \Phi_{max} \quad \forall p \in R$$

where Φ_{max} is the maximum admissible phase gradient set by the coherence bound ρ_{min} .

Proposition J.2 (Propagation speed collapse at saturation). The propagation speed of coherent phase modes is:

$$v_{prop} = 1/n(x) = 1/f(\rho_{strain})$$

As $|\nabla \Phi| \rightarrow \Phi_{max}$, the strain density $\rho_{strain} \rightarrow \infty$ and thus $v_{prop} \rightarrow 0$.

Definition J.3 (Causal Boundary). The causal boundary of a gradient-saturated region R is the hypersurface:

$$\partial R_{causal} = \{ p \in C \mid v_{prop}(p) = \varepsilon \}$$

for $\varepsilon \rightarrow 0^+$. This is the Phase Theory analogue of the event horizon.

Theorem J.4 (Singularity avoidance). The causal boundary ∂R_{causal} is a well-defined smooth hypersurface in C , and the phase configuration Φ remains continuous and bounded across it. Phase Theory therefore contains no geometric singularities —

the divergences of classical GR (Penrose–Hawking singularity theorems) do not arise because spacetime geometry is emergent rather than fundamental.

Proof sketch. The saturation condition $|\nabla\Phi| \rightarrow \Phi_{max}$ is a finite limit by the coherence bound $\rho_{min} > 0$. The configuration Φ itself remains in $L^\infty(C)$ throughout. The emergent metric $g_{\mu\nu}^{eff}$ degenerates at $\partial\mathcal{R}_{causal}$, but this degeneration is a coordinate artifact of the projection from C to emergent spacetime — not a divergence in the underlying substrate. $\square$

K. RECOVERY OF KNOWN PHYSICS

Phase Theory recovers established physical laws as effective descriptions in the appropriate limits.

K.1 — Quantum Mechanics (Sector quantization from topology)

The quantization of observables in quantum mechanics arises from the discreteness of homotopy groups $\pi_n(C)$. The winding number $k \in \mathbb{Z}$ is the topological precursor to quantized charge; the Hopf invariant Q_H gives rise to spin quantization. The superposition principle follows from the linearity of the phase-gradient energy in the weak-defect limit.

K.2 — Quantum Field Theory (Effective excitation algebra)

In the regime of many weakly interacting defects at low excitation index ι , the algebra of phase-mode creation and annihilation operators converges to the Fock-space algebra of QFT. The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges as the holonomy group G of the dominant stable defect sectors.

K.3 — General Relativity (Weak-strain limit of g_{eff})

In the limit $|\nabla\theta|^2 \ll 1$ and $|\nabla \times A|^2 \ll 1$, the transport metric $g_{\mu\nu}^{eff}$ satisfies the Einstein field equations:

$$G_{\mu\nu}[g^{eff}] = 8\pi G_N T_{\mu\nu}[\Phi]$$

where $G_{\mu\nu}$ is the Einstein tensor, G_N is Newton's constant (derived from $\alpha, \beta, \gamma, \delta$ — not inserted by hand), and $T_{\mu\nu}[\Phi]$ is the effective stress-energy of the phase configuration.

K.4 — Classical Optics (Coherent mode propagation)

In the absence of topological defects, $I[\Phi]$ is minimized by coherent propagating modes satisfying:

$$\nabla^2 \theta - n^2(x) \partial^2 \theta / \partial t^2 = 0$$

the wave equation with spatially varying refractive index — recovering classical optics and Maxwell's equations in vacuum.

L. FALSIFIABILITY CONDITIONS

Phase Theory is empirically *falsified* — not merely weakened — by any of the following observations:

L.1 — Causal Loop. Existence of any stable physical process realizing a closed causal loop:

$$\exists \text{ stable closed directed cycle in } G_u$$

This would contradict Theorem I.2 and the monotone-inconsistency-decrease law simultaneously.

L.2 — Mass-Conjugation Asymmetry. Observation of any mass difference between a particle and its conjugate:

$$m(k) \neq m(-k)$$

This would contradict Proposition F.1 and the reflection symmetry of I .

L.3 — Gravitational Matter-Antimatter Asymmetry. Direct measurement of a difference in gravitational acceleration between matter and antimatter defects, which would falsify Proposition F.2.

L.4 — Spectrum Overflow. Discovery of a stable new particle beyond the predicted spectrum bound of Theorem E.5 ($N_{max} \approx 25 \pm 10$), with quantum numbers incompatible with any label in the quintuple (k, Q_H, t, χ, R) .

L.5 — GR-Exact Lensing. Observation of gravitational lensing deflection angles exactly consistent with GR at all measurable impact parameters, with zero room for the predicted $O(|V^2 \theta|^2)$ refractive corrections (Theorem H.4). Absence of deviation at current precision does not falsify; zero deviation at arbitrarily high precision would.

CANONICAL STATEMENT OF PHASE THEORY

The Fundamental Variational Principle

$$\text{Reality} \equiv \arg \min_{\Phi} I[\Phi]$$

subject to topology and coherence.

All physical laws — quantum, relativistic, gravitational, optical — are asymptotic effective descriptions of this global minimization problem in appropriate limiting regimes. The universe is not built from spacetime and fields; it is the geometry of global phase consistency.

End of Appendix A. Cross-references: §I (Introduction), §IV (Particle classification), §VII (Spectrum bounds), §VIII (Annihilation), §IX (Falsifiable predictions).